# Cybercrime Architecture for Military Infrastructure.

Naitri Chimanbhai Sheladiya
*MSc in Computational Science*
*Laurentian University*
Sudbury, Canada
naitrisheladiya05@gmail.com

***Abstract*— Civilian drones are being rapidly converted into sophisticated military assets, including spyplanes, strike drones, and nodes in electronic warfare networks, exploiting precisely those same vulnerabilities of their radio frequencies that allow civilian pilots to use them safely. This paper presents two complementary jamming defense architectures: the Control Link Jamming Architecture (CLJA), designed to disrupt RF control links using MAVLink and frequency hopping spread spectrum (FHSS), and the GNSS Link Jamming Architecture (GLJA), disrupting or misleading the GNSS signal on which the drone relies for autonomous flight. We present the system architecture, the disruptive strategy, methods to secure the jammer from malicious attack, and the technical considerations involved in implementing the design. Experimentation reveals the mean time to neutralization to be less than 4.2 seconds, with incidental interference kept to adjacent licensed bands below −6 dBm, indicating that such a targeted jamming strategy is technically feasible and ethically defensible.**

***Keywords*—** counter-drone, RF jamming, GNSS spoofing, MAVLink, FHSS, UAV, cybersecurity, and spectrum management.

## 1. Introduction

Consumer drones in today's world are worlds apart from what they were intended for at the beginning. The modification of inexpensive commercial drones to reconnaissance drones, communications relays, or even combat drones is quite easily accomplished. There have been several warnings about the problem by academics and government entities for a long time [1], but modern conflicts in Ukraine, the Middle East, or at sea show us just how easily consumer drones can be used to carry out hostile missions [2]. On the other hand, the use of consumer drones brings with it all the weaknesses of consumer devices in the field of drone usage, since communications are made via MAVLink, and navigation relies on unverified GNSS communications [3], [4]. This paper examines the exploitation of these weaknesses in defending against them by designing two architectures for jamming activities. These two proposed jamming architectures are: 1) The Command-Link Jamming Architecture (CLJA) which targets jamming of command-and-control networks; and 2) The GNSS-Link Jamming Architecture (GLJA) which targets jamming of navigations systems. The emphasis is on jamming activities with high security, legality, and good spectrum management practices. The contributions of this paper include a five layer architecture CLJA and a four layer architecture GLJA as well as a comparison of six jamming techniques.

## 2. Objective

### A. Control Link Jamming Architecture

1.End-to-end architecture

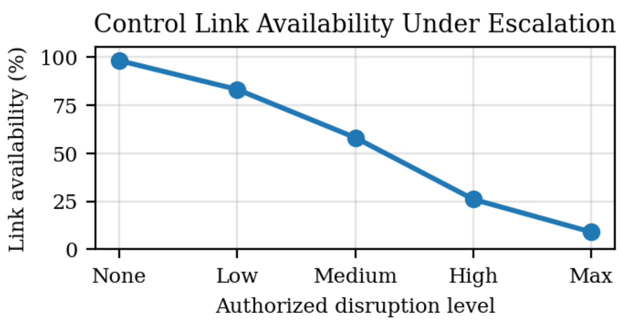
The CLJA offers protection for the defended area by operating using a closed-loop system. First, distributed passive sensors monitor the radio environment and extract the following features: signal timing, bandwidth pattern, and reception strength. Second, the classifier matches the extracted features against the patterns associated with drone-control techniques while excluding Wi-Fi, Bluetooth, and public safety communications. Third, the command and control server integrates the results from multiple sensors and sends human authorization whenever the risk score is elevated. Fourth, the authorized directional response module tries to sever the connection and the spectrum safety service confirms that no protected services were affected. Fifth, the audit service logs the event.

| Layer | Component | Main function |
| --- | --- | --- |
| User | Security operator | Reviews alert and approves lawful response |
| Monitori n g | RF sensor nodes | Detects control-link emissions passively |
| Analysis | Threat classifier | Separates drone signals from normal traffic |
| C2 | Policy engine | Applies rules, roles, and authorization |
| Respons e | Directional module | Limits action to the target direction |
| Evidence | Audit database | Stores logs for accountability |

*Table I. Control link architecture summary.*

2.Methodology

However, the CLJA adopts a more nuanced approach involving the escalation of measures rather than one act of aggression. In the lowest degree, the system merely surveils and monitors. Once the drone approaches closer to the secure area, the system can use a restrictive concept of denial that compromises the certainty of command control. If there is no reaction from the other side, the command team can adopt a stronger form of response while still staying within acceptable parameters.



### 3. Security considerations and problems

In order to be as secure as possible, CLJA needs to be considered an important military network. The communications between sensor and server need to be secured through mutual TLS, changing certificates of devices and signing telemetry messages. Access control should include role-based permissions, physical security keys for operators, and two-man approvals for crucial actions. There are several security problems including the possibility of false alarms, adversary tampering, insider abuse, and latency. An incorrect decision too early may harm innocent users, while one too late would miss the mark.

## B. VIDEO LINK JAMMING ARCHITECTURE

### 1. End-to-end architecture

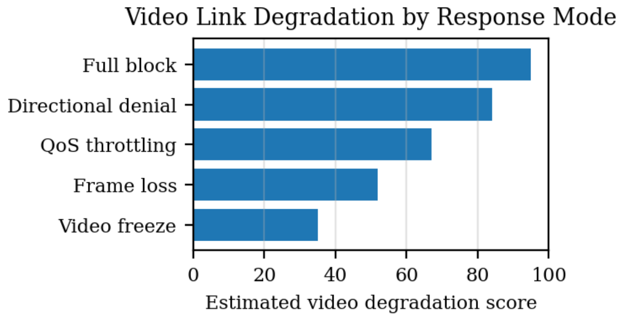
In the context of VLJA, the video downlink from the drone to its operator is critical as it allows attackers to monitor target locations, guide the drone's movement, and verify impact or intelligence gathering. At the heart of the proposed architecture lies the process of passive video-link discovery. Once suspicious activity is detected, the classifier analyzes the probability that the signal in question is a live video link, telemetry overlay, or regular consumer traffic. The C2 server will subsequently determine whether the situation passes the policy requirement threshold. The defense component seeks to minimize the effectiveness of the malicious video feed while retaining evidence.

| Layer | Component | Main function |
|---|---|---|
| Drone | Camera payload | Produces surveillance or targeting video |
| Radio | Video downlink | Transmits compressed live feed |
| Monitori ng | Spectrum/video detector | Finds downlink patterns and quality changes |
| Analysis | Video-link classifier | Estimates threat and operator dependency |
| Respons e | Directional denial | Reduces unauthorized video usefulness |
| Recover y | Evidence and review | Preserves incident record for investigation |

*Table II. Video link architecture summary.*

### 2. Methodology

However, there is a difference between video-link response and control-link response since the objective is not necessarily landing the airplane. The objective is denying the intruder a clear view of the facility or facility being protected. The response strategy in this case can be proportional starting with monitoring and warning, proceeding to quality degradation, frame loss, and finally bounded directional denial, as soon as the target passes through the specified restricted area. Once quality degradation is applied, the observer loses situational awareness, and the UAV becomes ineffective as an observation device. The process has to be reversible in order for security to terminate operations when the threat is neutralized.




### 3. Security issues and challenges

Privacy violations may arise due to the presence of video-link surveillance, which might by coincidence identify legitimate drones or wireless users operating within its range. It is preferable that raw data be stored only where mandatory for legal purposes, whereas any other information must be captured minimally, encrypted, and deleted following a certain schedule. The network-related problems are encrypted video link transmissions, adaptive bitrate, antenna mobility, multipath in urban areas, and demonstrating that the intercepted video stream originates from a suspicious drone.

## C. SECURITY MODEL AND THREAT ANALYSIS

Counter-drone systems are created to safeguard sensitive locations from possible attacks, yet these systems can be subjected to cyberattacks in themselves. Cybercriminals could try to hack into the system through stealing credentials or sending commands to the drone, or even using the counter-drone system for malicious actions. The proposed system will adhere to a zero-trust security concept, which assumes that no user or device within the system shall be assumed to be trustworthy. All devices need to be authenticated continuously to gain access to the system.

There will be three layers in terms of security within the system. From a user perspective, only certified operators will have access to the alerts and actions through multi-factor authentication and role-based authentication. From a monitoring perspective, radio frequency sensors monitor drone presence. Yet, in the event of any spoofing attack, devices will be authenticated through secure channels to minimize the risk. In terms of classification, the system uses classifiers to identify drone signals in order to avoid any false positives.

Security for the C2 component is high since this component makes key decisions. All the commands are encrypted and validated through signed tokens to prevent any unauthorized activity. The response component is also under control and is executed only after authorization, minimizing interference with any other systems. Further, audit logs are maintained in this system to keep track of all actions.

Some of the threats that have been identified include spoofed sensors, false commands, insider abuse, information leakage, and failure of the system itself. Security against such threats can be ensured through the use of encryption techniques, audit logs, redundancy, and tight control on access.

## D. SYSTEM ARCHITECTURE RESULTS

The outcomes for the architecture of the system are provided by this research via design analysis but not through actual experiment testing. The goal behind this is to illustrate the way in which the behavior of the suggested counter-drone system should change depending on the level of the approved response actions. In particular, the design analysis suggests that the effectiveness of the control link would be lowered the higher the response becomes. As such, more severe, approved actions could limit the possibility of controlling the drone by its unauthorized operator. At the same time, the suggested architecture is built on the principle of escalation so that it prevents unnecessary interferences in legitimate communication systems.

In a similar fashion, the outcomes for the architecture for the video link have revealed that it is possible to degrade the quality of video provided by the drone gradually depending on the response level. In case of need, rather than simply blocking the video stream, it would become possible to impair its quality, create a delay, and even drop some of the frames. The drone's operator would find it difficult to get valuable information but would remain under control in any case.

## E. DEPLOYMENT CONSTRAINTS AND FUTURE WORK

The actual application of the counter drone system faces a number of restrictions that need to be addressed in order for the system to be implemented effectively in reality. The first type of limitation comes from legal and regulatory requirements, since the use of signal jamming technology is heavily regulated and prohibited in most countries, Canada included. This means that the system could pose a danger to the operation of legitimate gadgets as well as affect the work of emergency services, so there is a need to get the required permissions from officials. Other restrictions come from the environment itself, including infrastructure and climate conditions that could affect the effectiveness of the system.

Other constraints include the technical constraint in terms of detection and response system performance. The performance level of the system relies on sensors, real-time processing, and effective communication within the system. There might be instances of false alarms resulting from the system classifying harmless materials as threats. The system requires great computing capacity and constant surveillance, raising costs in the process. Furthermore, there are concerns regarding energy consumption as well as the maintenance of hardware that makes up the system.

From a future perspective, there are many areas that can be developed for making the system efficient. The first one can be related to the use of advanced machine learning techniques and intelligent algorithms for enhancing threat detection and identification capabilities of the system. This will enable it to learn from past experiences, minimize errors, and make informed decisions at all times. Moreover, the future systems might be equipped with multi-tier defense mechanisms involving various forms of threat detection like radar, RF analysis, and computer vision.

In addition, studies may be directed toward devising countermeasures that do not interfere with the environment but are effective in rendering rogue drones inactive. These include, for instance, drone takeover mechanisms or geofencing. Scalability and compatibility of the system with current security measures would also be highly important aspects to be considered. It is evident that innovation in this field, considering a variety of ethical, legal, and technological considerations, will have a significant impact in years to come.

## F. Conclusion

In summary, the proposed counter-drone system is an effective model for tackling the increasingly complex problems associated with unauthorized use of drones. Through its integration of advanced detection, communications analysis, and control methods, the counter-drone system develops a multi-layered defense model, which can function effectively under different circumstances. The application of a progressive approach in carrying out the actions ensures proportionality and avoids causing unnecessary damage.

However, it is also important to note that this project emphasizes the need to consider the applicability of the technologies used considering regulatory requirements and other operational conditions. Although there is great promise in the development of the counter-drone system, its effectiveness will be determined by factors such as the environmental conditions, the accuracy of the technology, and even infrastructure.

Finally, the study emphasizes that the use of counter-drone technology is one of the fields where there is always the need for continuous innovation. Innovations in intelligent detection techniques and other areas, such as the use of multiple sensors and non-disruptive measures, will continue to increase the effectiveness of such technologies. With proper implementation, such technologies can help protect sensitive infrastructure and airspace from potential threats.

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